Web Design INSY8314

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Transform Your Store with Smart Inventory Management

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StoreMinder Inventory Management System

Web Design Final Exam Project Documentation

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1 Project Introduction

1.1 Overview

Title: StoreMinder - Modern Inventory Management System

Case Study: StoreMinder is designed for small to medium-sized retail businesses, hardware stores, and quincaillerie shops in Rwanda. It empowers business owners and store managers to efficiently manage inventory, track sales, generate reports, and handle customer transactions.

Purpose: StoreMinder addresses the challenges of manual inventory management by providing:

- Automated inventory tracking and stock management
- Real-time sales recording and transaction management
- Comprehensive business reports and analytics
- Multi-user access with role-based permissions
- Data security and backup for business continuity
- **Technologies Used:**
- **Frontend:** HTML5, CSS3, JavaScript (ES6+)
- **Backend:** PHP 8.0+
- **Database:** MySQL 8.0+
- **Server:** XAMPP (Apache + MySQL + PHP)
- **Additional Libraries:**
  - Chart.js for data visualization
  - FPDF for PDF report generation
  - PHPMailer for email functionality
  - Simple Icons for UI elements

2 Problem Statement

Manual inventory management in small businesses is:

- Time-consuming and error-prone

- Difficult for accurate stock tracking
- Challenging for generating business insights
- Vulnerable to data loss
- Limited in scalability

Target Users:

1. **Business Owners:** Need business performance and financial reports
2. **Store Managers:** Require daily inventory and sales tracking
3. **Sales Staff:** Need to record transactions and check product availability
4. **Administrators:** Manage user accounts and system settings

Key Features:

- Real-time inventory tracking with low-stock alerts
- Sales recording with multiple payment methods
- Customer transaction management
- Comprehensive reporting and analytics
- Multi-user access with role-based security
- Responsive design for mobile and desktop

3 System Requirements

3.1 Software Requirements

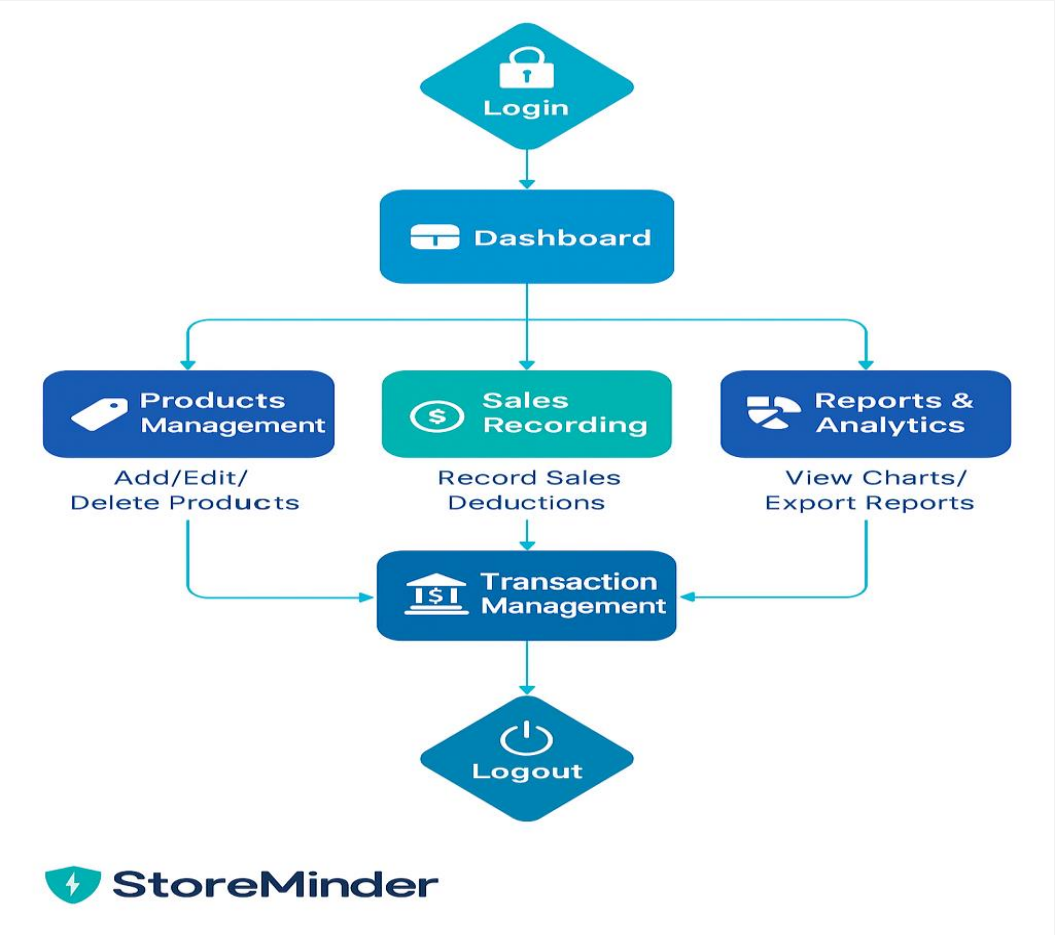
- **Web Browser:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **Code Editor:** Visual Studio Code (recommended)
- **Server Environment:** XAMPP 8.0+ (Apache + MySQL + PHP)
- **Database:** MySQL 8.0+
- **PHP Version:** 8.0 or higher

3.2 Hardware Requirements

- **Minimum:** 4GB RAM, 2GB free disk space
- **Recommended:** 8GB RAM, 5GB free disk space
- **Network:** Internet connection for external libraries and email functionality

4 System Design

4.1 User Flow Diagram



4.2 Page Structure

1. index.php - Landing page with login/signup modals
2. dashboard.php - Business overview dashboard
3. products.php - Product inventory management
4. sales.php - Sales recording and deduction tracking
5. transactions.php - Customer transaction management

6. transaction*receipt.php−Receiptgenerationandsharing*reports.php−*Businessanalyticsandrep*

7.7. users.php - User management (admin only)

8. login.php - User authentication

9. register.php - User registration

4.3 Navigation System

- Fixed header with responsive navigation menu
- Role-based menu items
- Breadcrumb navigation
- Mobile-responsive hamburger menu

5 Implementation

5.1 HTML Structure

- Semantic tags: <header>, <nav>, <main>, <footer>, <section>, <article>
- Clean, well-indented code
- Accessible form labels and ARIA attributes

Responsive meta tags and viewport settings

Modal dialogs, data tables, form validation attributes, responsive grid layouts

5.2 CSS Styling & Responsiveness

- CSS Variables for consistent theming
- Modern gradient backgrounds and glassmorphism effects
- Mobile-first responsive design
- Professional color scheme and typography
- CSS Grid, Flexbox, media queries, smooth transitions

5.3 JavaScript Interactivity

- Theme toggle (dark/light mode)
- Modal open/close functionality
- Form auto-fill and validation
- Smooth scrolling and animations
- Dynamic content loading

5.4 JavaScript Form Validation

- Real-time email format validation
- Required field checking
- Password strength validation
- Form submission prevention on errors
- User-friendly error messages

5.5 PHP Form Handling

- Secure POST method handling
- Comprehensive error handling
- Session management and security
- File upload handling for receipts
- Email functionality integration

5.6 Database Integration

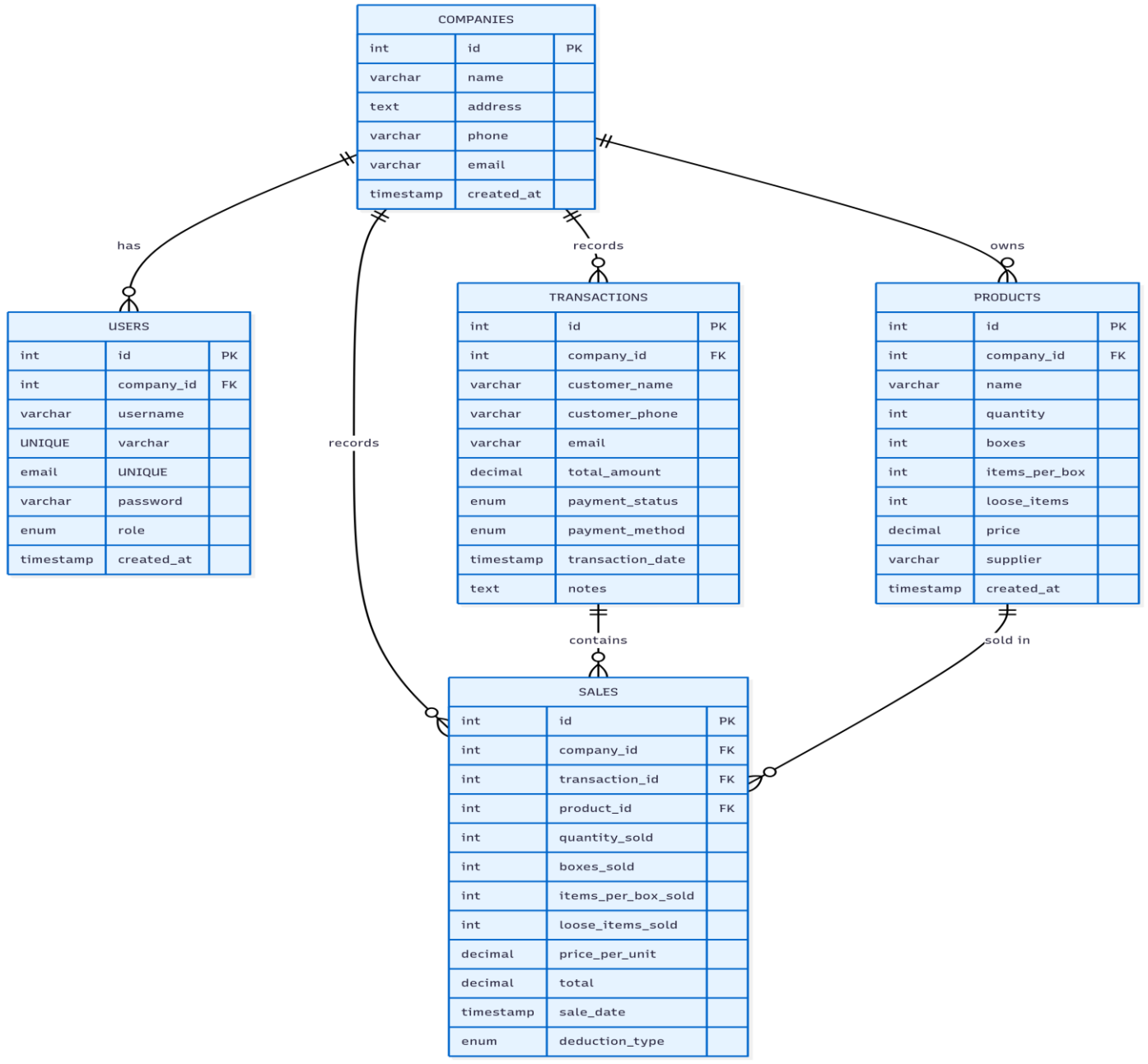
- Proper database relationships and foreign keys
- Optimized queries with prepared statements
- Transaction support for data integrity
- Backup and recovery procedures

6 Database Design

6.1 Database Name

quincaillerie$_{m}anager$

6.2 ER Diagram Structure



6.3 Tables and Relationships

6.3.1 companies

- id (INT, Primary Key, Auto Increment)
- name (VARCHAR(255), NOT NULL)

- address (TEXT)
- phone (VARCHAR(20))
- email (VARCHAR(100))
- created$_{a}t$(*TIMESTAMP*,*DEFAULTCURRENT$_{T}$IMESTAMP*)

6.3.2 users

- id (INT, Primary Key, Auto Increment)
- company$_{i}d$(*INT*,*ForeignKey*→ companies.id)
- username (VARCHAR(50), UNIQUE, NOT NULL)
- email (VARCHAR(100), UNIQUE, NOT NULL)

  password (VARCHAR(255), NOT NULL)

  role (ENUM(’admin’, ’user’), DEFAULT

  ’user’)
- created$_{a}t$(*TIMESTAMP*,*DEFAULTCURRENT$_{T}$IMESTAMP*)

6.3.3 products

- id (INT, Primary Key, Auto Increment)
- company$_{i}d$(*INT*,*ForeignKey*→ companies.id)
- name (VARCHAR(255), NOT NULL)
- quantity (INT, DEFAULT 0)
- boxes (INT, DEFAULT 0)
- items$_{p}er_{b}ox$(*INT*,*DEFAULT*0)

6.3.4 transactions

- id (INT, Primary Key, Auto Increment)
- company$_{i}d$(*INT*,*ForeignKey*→ companies.id)

- customer$_{n}ame(VARCHAR(255), NOTNULL)$

6.3.5 sales

- id (INT, Primary Key, Auto Increment)
- company$_{i}d(INT, ForeignKey \rightarrow$ companies.id)
- transaction$_{i}d(INT, ForeignKey \rightarrow$ transactions.id)
- product$_{i}d(INT, ForeignKey \rightarrow$ products.id)
- quantity$_{s}old(INT, NOTNULL)$

6.4 Key Relationships

- Multi-tenant architecture with company isolation
- Referential integrity with CASCADE delete
- Optimized indexes for performance

7 Testing

7.1 Testing Methodology

1. **Manual Testing:**
    - User registration and login
    - Product CRUD operations
    - Sales recording and transaction management
    - Report generation and export
    - Role-based access control
2. **Form Validation Testing:**
    - Client-side (JavaScript)
    - Server-side (PHP)
    - Database constraints • Error handling and feedback

3. **Cross-browser Testing:**
    - Chrome, Firefox, Safari, Edge compatibility
    - Mobile responsiveness • Different screen sizes
4. **Security Testing:**
    - SQL injection prevention
    - XSS protection
    - Session security
    - Input sanitization

7.2 Bugs Found and Fixed

1. **Foreign Key Constraint Error:** Fixed missing companies table relationships
2. **Missing Database Column:** Added loose*itemstoproductstable***Navigation Overlap:***Fixedth*
3. **Database Connection Issues:** Implemented proper error handling

8 Challenges Faced

1. **Multi-tenant Architecture:**
    - *Challenge:* Implementing company isolation
    - *Solution:* Used foreign key relationships and company-based filtering
2. **Database Schema Evolution:**
    - *Challenge:* Adding columns without data loss
    - *Solution:* Used ALTER TABLE with default values
3. **Responsive Design:**
    - *Challenge:* Consistent UI across screen sizes
    - *Solution:* Mobile-first CSS with Grid and Flexbox
4. **Security Implementation:**

- *Challenge:* Preventing SQL injection
- *Solution:* Used prepared statements and sanitization

5. **User Experience:**
   - *Challenge:* Intuitive interface for non-technical users
   - *Solution:* Modern UI patterns and clear navigation

9 Conclusion

9.1 Learning Outcomes

1. **Full-Stack Development:** Expertise in HTML, CSS, JavaScript, PHP, MySQL
2. **Database Design:** Normalization and relationship management
3. **Security Best Practices:** Secure coding and data protection
4. **User Experience Design:** Intuitive and responsive interfaces
5. **Project Management:** Planning, implementation, and testing

9.2 Future Improvements

1. Advanced analytics with machine learning
2. Native mobile applications
3. RESTful API development
4. Cloud infrastructure migration
5. Sophisticated reporting tools
6. Barcode scanning and automated stock alerts

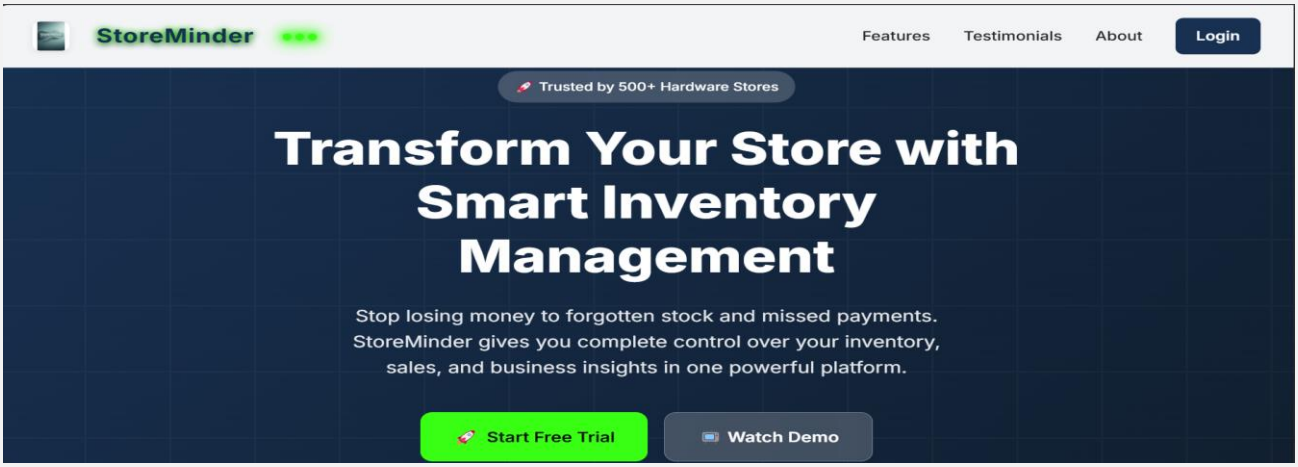
9.3 Project Success Metrics

- Complete inventory management functionality
- Multi-user role-based access control
- Responsive design for all devices

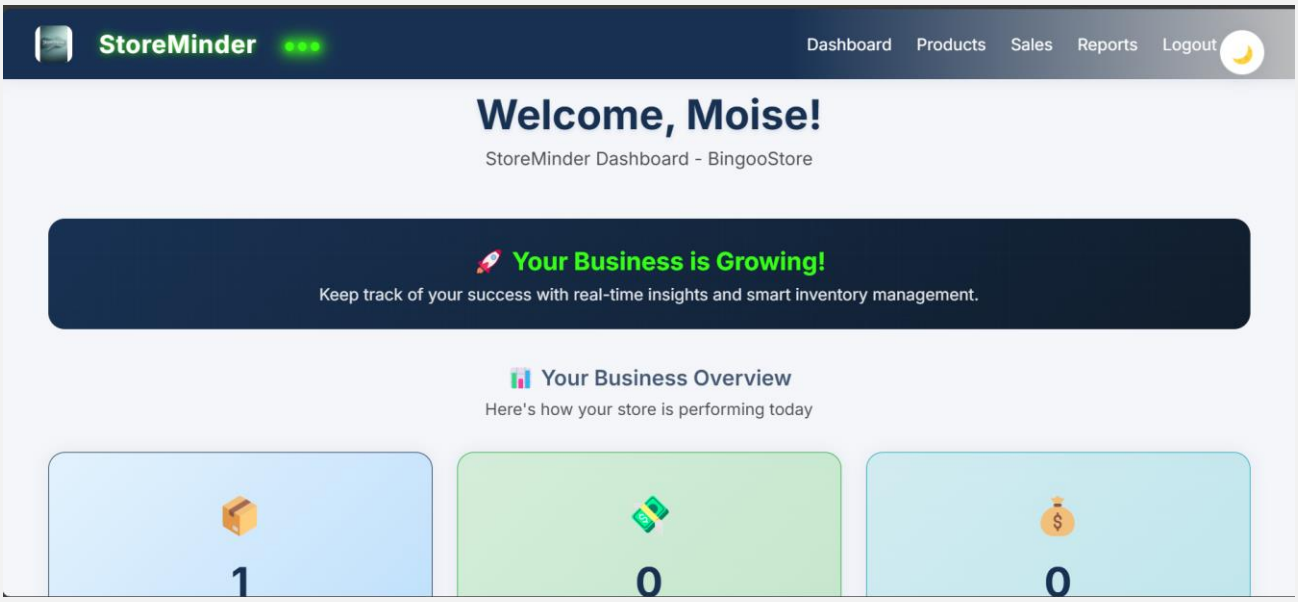
- Comprehensive reporting system
- Secure data handling and storage
- Professional UI/UX design

10 Screenshots

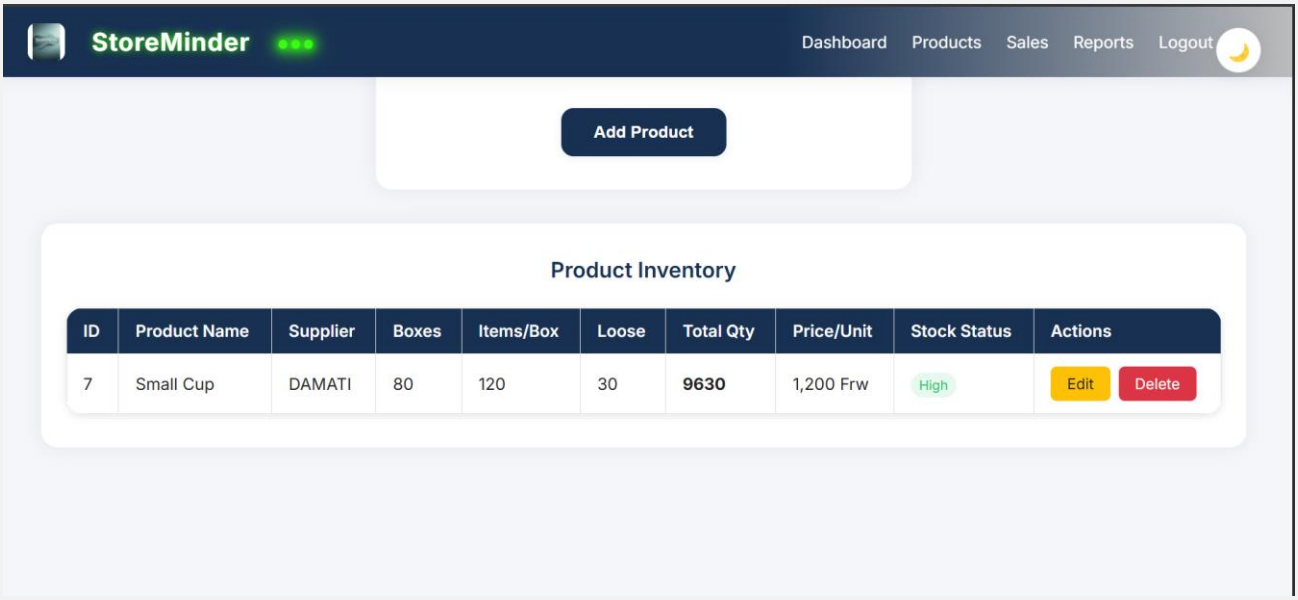
1. **Landing Page (index.php):** Hero section, login/signup modals, mobile view



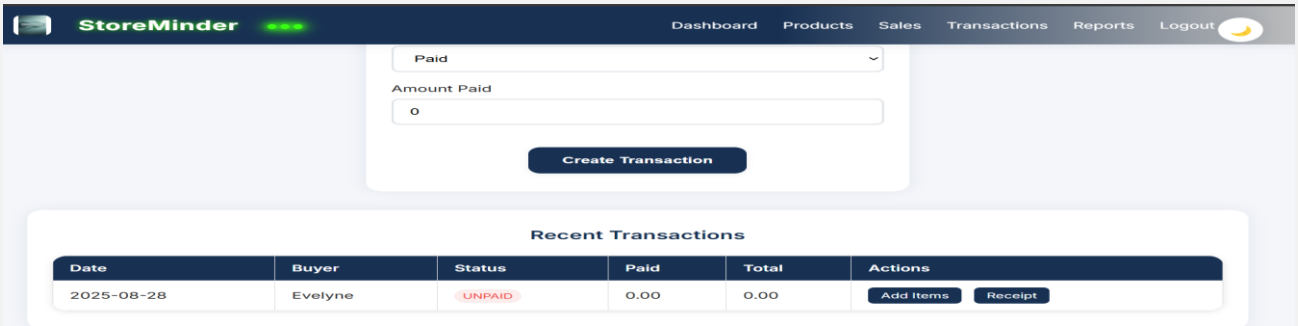
2. **Dashboard (dashboard.php):** Business overview, quick actions, low stock alerts



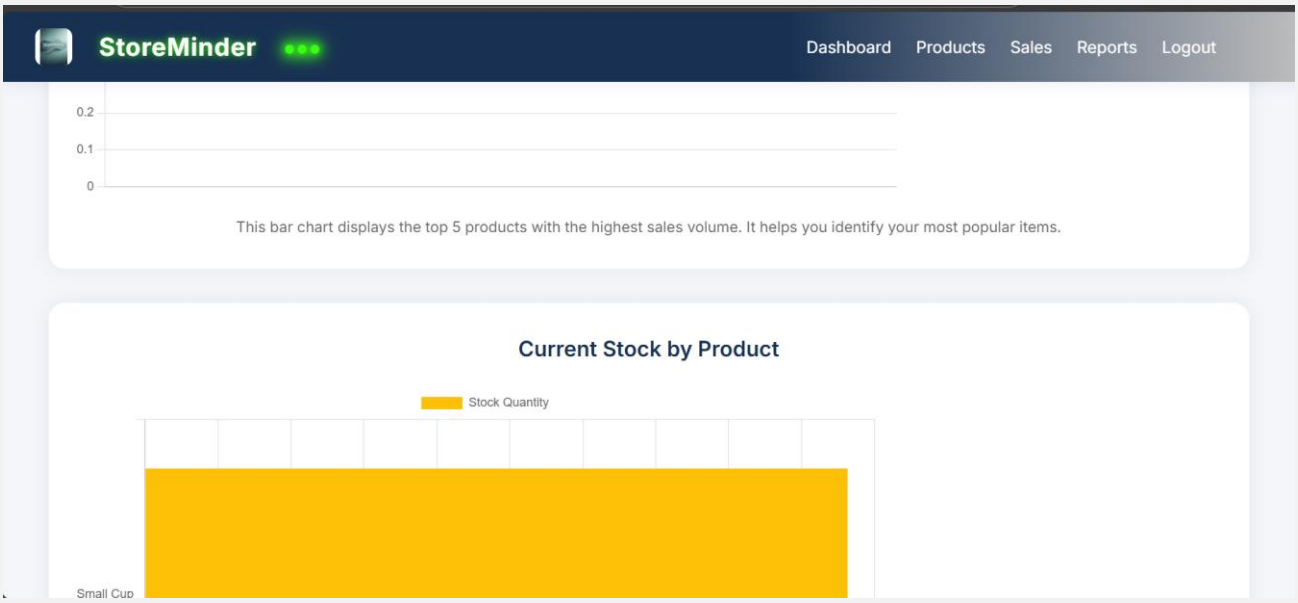
3. **Products Management (products.php):** Add product form, inventory table, edit/delete



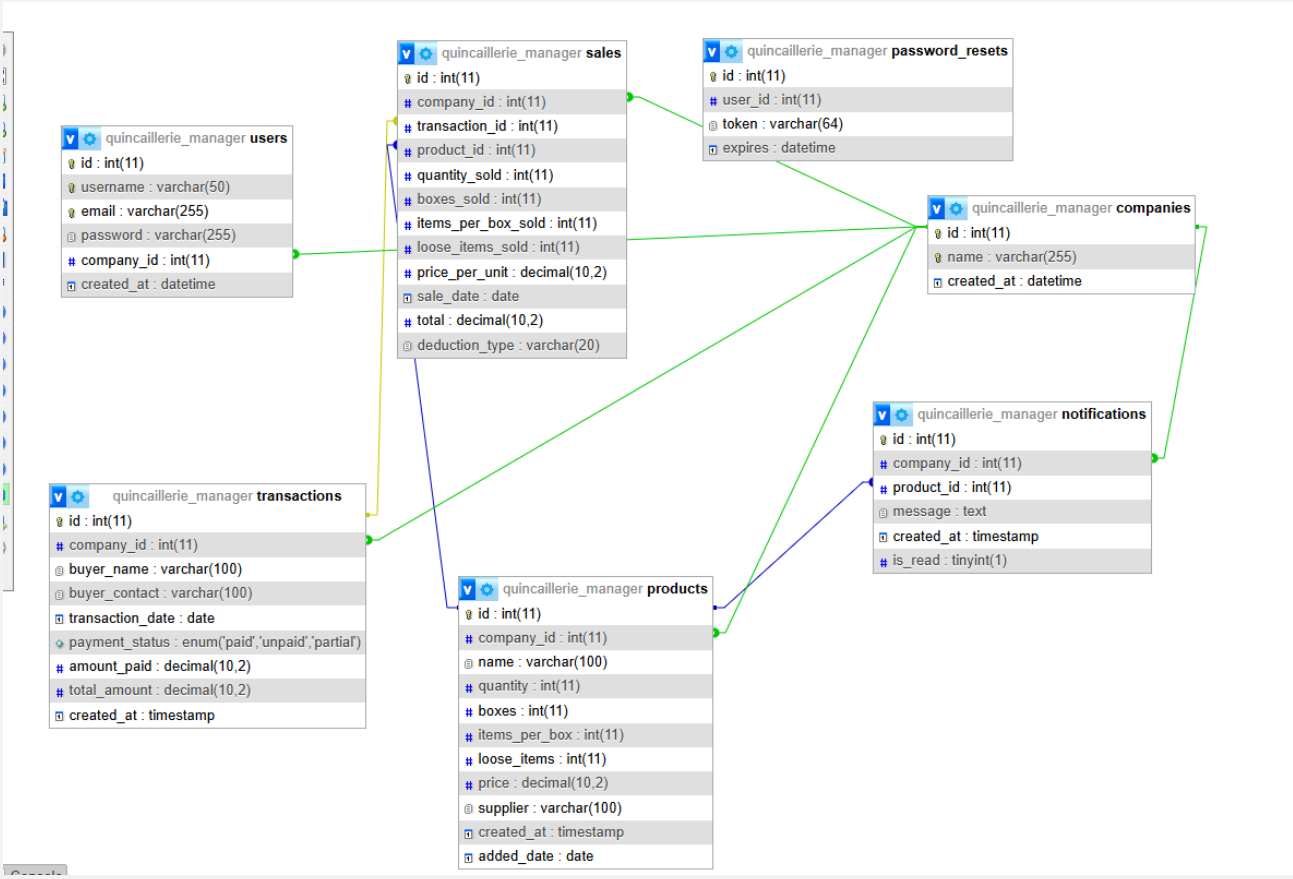
4. **Sales Recording (sales.php):** Sales form, history table, transaction details



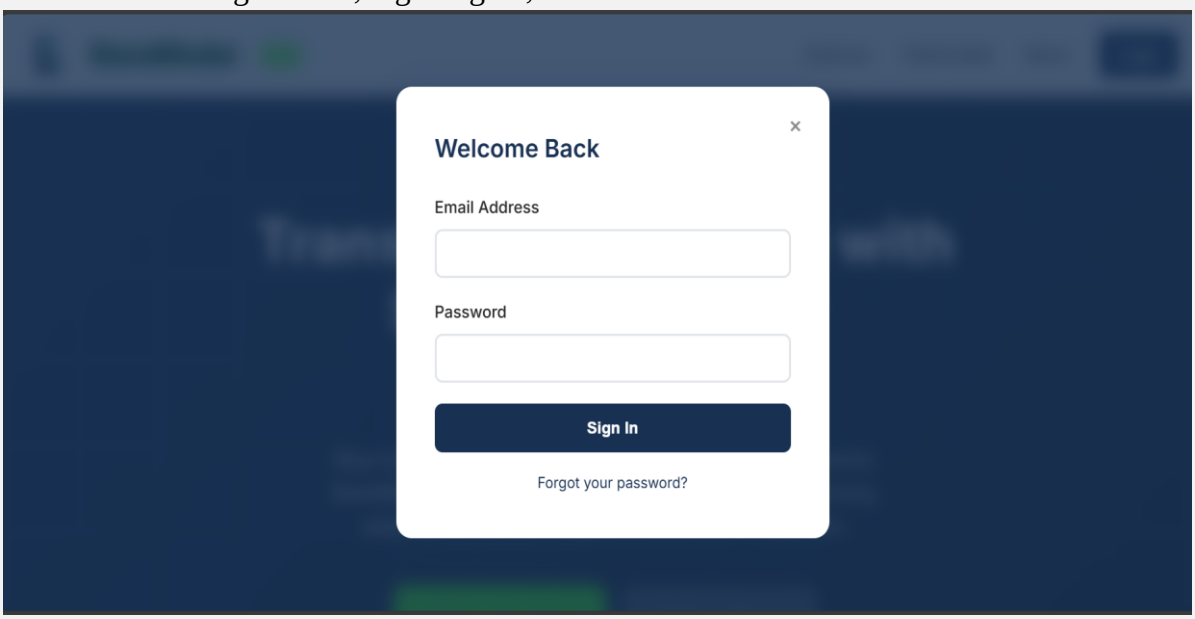
5. **Reports (reports.php):** Charts, export functionality, filter options

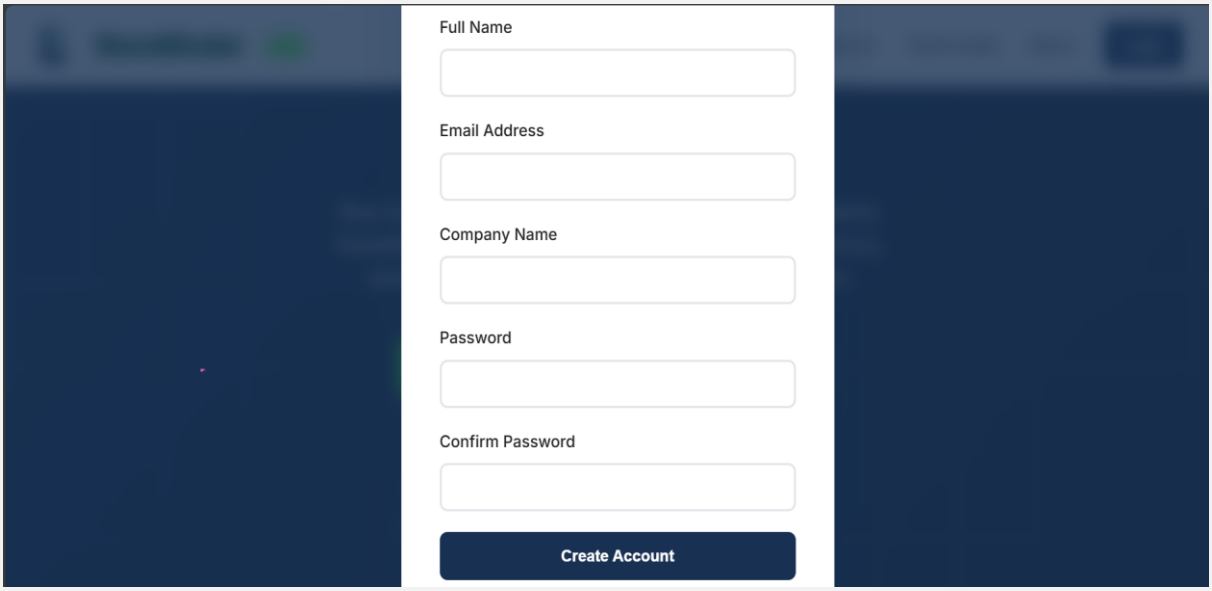


6. **Database Views:** phpMyAdmin table structure, sample data, foreign keys



7. **User Actions:** Registration, login/logout, role-based access



8. **User SignUp**

StoreMinder - Inventory Management System